

MUHAMMAD USMAN BARI

AI/ML Learner | Full-Stack Developer

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PROFESSIONAL SUMMARY

Computer Science undergraduate at FAST-NUCES with a strong foundation in software engineering, full-stack development, and modern web technologies. Currently expanding expertise in Artificial Intelligence and Machine Learning through hands-on projects involving Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), and intelligent question-answering systems. Skilled in React, Next.js, TypeScript, Python, and building end-to-end applications. Passionate about developing AI-driven solutions and applying machine learning techniques to solve real-world problems.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++

AI & ML: LLMs (OpenAI API, Gemini API, Groq API), RAG pipelines, LangChain-style architectures, Vector databases, prompt engineering, NLP (tokenization, semantic search, embeddings), HuggingFace Transformers

Frontend: React.js, Next.js, HTML, CSS, Tailwind CSS

Backend & Databases: Node.js, Express.js, Firebase, MySQL, REST APIs

Tools & Platforms: Git, GitHub, Postman, Linux, Visual Studio Code

Soft Skills: Problem Solving, Team Collaboration, Communication, Critical Thinking, Time Management

EDUCATION

BS Computer Science | FAST-NUCES, Islamabad | *Aug 2023 – Present*

PROFESSIONAL EXPERIENCE

Web Development Intern | CodeAlpha | *July 2025 – Aug 2025*

- Integrated LLM APIs into production React/Next.js interfaces, building real-time AI response streaming and conversation state management.
- Developed UI components for AI-powered features, collaborating with a team under Git-based workflows and code review processes.
- Built reusable components and optimized application performance, applying clean code principles and component architecture to ship AI feature interfaces on deadline.
- Strengthened understanding of software engineering principles and user-centered design through professional team collaboration.

CERTIFICATIONS & ACHIEVEMENTS

COURSERA | META | Front End Development Specialization | *Apr 2025*

COURSERA | Google | Data Analytics Certification | *May 2025*

PROJECTS

Enterprise RAG Knowledge Assistant | *Python, Groq API, FAISS, LangChain*

- Built a production-style RAG pipeline that answers natural-language questions over large PDF corpora (200+ pages) with context-aware, cited responses.
- Engineered a chunking + embedding pipeline (text-embedding-ada-002) feeding a FAISS vector store; reduced irrelevant retrievals by ~35% via re-ranking strategies.
- Optimized prompt templates to cut hallucination rate and improve answer grounding — validated across 50+ diverse test queries.
- Designed end-to-end document ingestion pipeline covering chunking, embedding generation, and vector-based semantic search.

QueryMind – AI-Powered Data Assistant | *Python, Groq API, MySQL, SQLAlchemy*

- Built an NL-to-SQL engine that parses plain-English questions into executable SQL — achieving ~90% query accuracy on a 10-table relational schema.
- Auto-generates ERDs from live database schemas, enabling non-technical users to explore and query data without writing SQL.
- Handles JOIN inference, aggregations, and nested queries via few-shot prompting combined with schema injection into the LLM context.
- Streamlined data querying workflow, significantly reducing dependency on manual SQL writing for end users.

Intelligent Reading Comprehension System | *Python, NLP, HuggingFace Transformers*

- Developed an NLP-based question-answering system capable of extracting contextual answers from textual passages using transformer-based models.
- Implemented text preprocessing, tokenization, and information extraction techniques for structured language processing.
- Improved comprehension and retrieval accuracy through optimized language processing pipelines.

AI Chatbot Application | *React, Next.js, TypeScript, Tailwind CSS, Gemini API*

- Built an AI-powered chatbot with real-time conversational capabilities, integrating Gemini API for intelligent response generation.
- Developed reusable UI components and modern frontend architecture with responsive user experience.

Socially – Full Stack Social Media Application | *Android, Firebase, REST APIs, MySQL*

- Developed a feature-rich social networking platform supporting authentication, messaging, media sharing, and user interactions.
- Implemented backend integrations and optimized data synchronization across application components.

SmartBus Transport System | *Java, SQL*

- Designed and developed a transport management system for route scheduling, bus allocation, and passenger management.
- Implemented database integration and efficient data handling using SQL